

Discussion: “Carbon-Adjusted Fiscal Multiplier” by Henrique Basso, O. Rachedi & V. Nispi Landi¹

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General impression

- Research Question combines two important & timely issues:
 - fiscal policy
 - climate externalities
- Paper is clear about its contribution and highly polished
- Battery of robustness checks in the Appendix
- A pleasure to read!

Part I: Recap

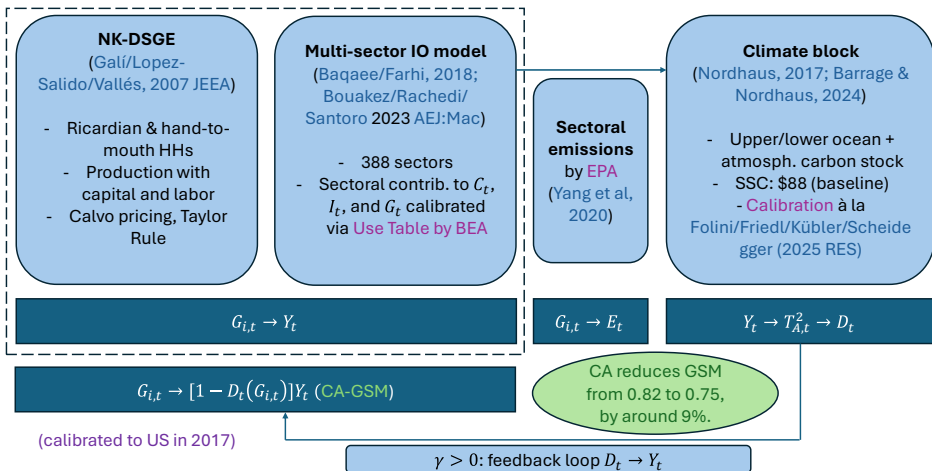
Recap of the paper

Research question: “How are government spending multipliers diminished if one takes into account their effect on greenhouse gas emissions?”

Structural model combining different elements which have not been joint so far:

- government spending multiplier (GSM)
- input-output model for heterogeneous carbon intensity across sectors
- damage function by greenhouse emissions

Recap: Carbon-Adjusted Gov. Spending Multiplier



Part II: Comments

1. The CA-GSM depends on country size (!)

Central: damage function mapping $\Delta^\circ\text{C}$ into %-GDP loss where
“**climate damages increase with the square of atmospheric temperature anomaly**”:

$$1 - D_t = 1 - \gamma T_{A,t}^2 \quad (33)$$

This implies (roughly):

Cty	Y_t	ΔG_t	$\Delta T_{A,t}$	$\Delta T_{A,t}^2$	$\Delta_{CA} \text{ GSM}$
US	\$20trn	\$20bn	0.001°C	10^{-6}	8.5%
DE	\$4trn	\$4bn	0.0002°C	4×10^{-8}	0.21%
PT	\$0.2trn	\$0.2bn	0.00001°C	10^{-10}	0.0009%

⇒ Smaller countries: **negligible % damages and minimal CA-GSM!**

2. Short vs long run

“While the stock of atmospheric carbon peaks within one year of the public spending shock, the temperature response (...) takes roughly 10 years to reach its maximum [and] declines only gradually, **with a half-life well exceeding 500 years.** (...)”

“[A] **divergence** [in GDP paths with and without damages] **emerges roughly a decade after the shock**, as rising temperatures begin to erode productivity.”

Spending multiplier discounts to ∞ :

$$\mathcal{M} = \frac{\sum_{j=0}^{\infty} \beta^j \mathbb{E}_t (Y_{t+j} - \bar{Y})}{\sum_{j=0}^{\infty} \beta^j \mathbb{E}_t \left[\sum_{i=1}^{\mathcal{I}} \left(\frac{P_{i,t+j}}{P_{t+j}} G_{i,t+j} - \frac{\bar{P}_i}{\bar{P}} \bar{G}_i \right) \right]}$$

2a) Model lacks growth and gov. investment

Intro: “Our model captures both the short-run transmission of fiscal policy (...) as well as the long-run climate effects (...).”

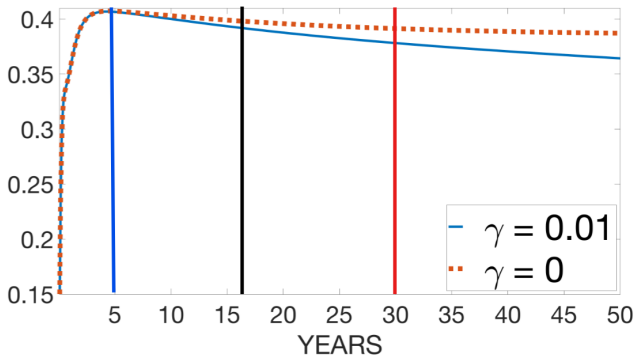
But:

- Model has no role for growth
- Model has no role for gov. investment I_t^G
 - US: Gross gov. investment I_t^G is about 25% of gov. consumption expenditures
 - Gov. investment can enhance TFP ([Antolin-Diaz and Surico, 2025](#))

⇒ Fiscal policy has (detrimental) long-run climate effects, but is not allowed to have (potentially beneficial) growth effects.

2b) What are political economy implications?

(f) GDP (Cumulative)



What would CA-GSM look like under discounting...

- ... that neglects effects after 4-5 years (election cycle)?
- ... that neglects effects after 25-50 yrs (one/two generations)?

3. Minor points

- All sectors suffer the same from climate damages – production function is $Z_{i,t} = F(N_{i,t}, K_{i,t}, H_{i,t}; D_t)$
 $\Rightarrow$ appropriate?
- Persistence of gov. spending is calibrated to $\rho^m = 0.7$.
 - implies quarterly persistence $\rho^q = (\rho^m)^3 = 0.34$.
 - Compare to US posterior of $\rho^q = 0.97$ in [Leeper, Plante, and Traum \(2010\)](#)!
- SCC: Paper assumes 1% GDP loss for 1°C, while NGFS now puts estimates of damages to 15% of GDP for 2°C by 2050.
<https://carbontracker.org/ngfs-scenarios-and-the-damage-done>

Conclusion

- Insightful paper with a very relevant research question
- Careful combination of the various required model parts
- Some caveats can potentially be acknowledged
- High publication potential!

Appendix

Minor comments

- You could repeat that persistence of the gov. spend. shock ($\rho = 0.7$) at the beginning of Section 4, as this is quite crucial.
- Does your IO model also have prices of productive inputs (R_t^K , W_t) that differ across sectors?
- robustness check: You could check robustness to results for higher values of elast. of subst. between sectors for capital and labor, $\nu_K, \nu_N \rightarrow \infty$ by putting large numbers here. I like the idea of imperfect mobility of the factors and suppose that is well motivated by [Bouakez, Rachedi, and Santoro \(2023, 2025\)](#). Still, a robustness check could be helpful, as not every multi-sector is using this friction.
- footnote 9: “Since our economy operates at a monthly frequency, compared to their annual framework, all coefficients are divided by 12. The same applies to the parameters of the temperature module.”
 $\Rightarrow$ unclear to me which of the parameters are divided by 12.

■ Possible extensions (future research?)

- Make the damage function D_t differ across countries and sectors.
- Model focuses on government spending.
 - How would tax multipliers in the model be affected by the carbon adjustment?
 - What about considering distortionary taxation (say, on consumption, labor or capital) instead of lump-sum taxes?
- How to explain the reduction in carbon footprint of government spending found in [Bucchianico, di Serio, Fragetta, and Melina \(2025\)](#)?

■ Typos

(virtually nonexistent!)

- footnote 3: “While our approach parsimoniously capture...”
- “Given the atmospheric temperature anomaly, environmental damages equal to”
- “The table reports the fraction of the GDP response that happen”
- eq (A.2): “ i ” index missing on LHS
- carbon reservoir calibration (before footnote 9) uses φ , while model equivalent uses ϑ
- “Using this metrics”
- the last two paragraphs in Section 3 repeat each other
- Section 4.1 first sentence “To illustrate how public spending affects emissions, temperatures, and climate damages in our model (...)” repeats sentence in paragraph above.
- “it takes few decades to dissipate half of the original emission pulse.” → “several”?
- footnote 12: “A positive adjustment emerges when public spending (...) (ii) crowds out output in carbon-intensive sectors.” → “crowds out private consumption / private demand” (?)
- “atmospheric carbon concentration peaks after eight months and takes nearly 50 years to halve” vs “Thereafter, the temperature anomaly declines only gradually, with a half-life well exceeding 500 years.”
- time index missing in damage function eq. (33)?

References I

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- Bouakez, H., O. Rachedi, and E. Santoro (2023, January). The government spending multiplier in a multisector economy. *American Economic Journal: Macroeconomics* 15(1), 209–39.
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